Master Specialization Proposal: Comprehensive Climate, Glacial, and Watershed Analysis

This proposal details a massive, unified academic implementation plan merging the objectives of Climate Change, Glacial Retreat, and Watershed Modeling analysis. It leverages **ArcGIS Pro (ArcHydro), ENVI, RStudio, and pure Python environments** to create a robust analytical pipeline.

Furthermore, to elevate the project to an enterprise-grade Geo-AI architecture, the specialization progresses into advanced topics including **Agentic Orchestration (LangGraph), Multi-Task PyTorch, Multi-Temporal Change Detection, PostGIS, and Full-Stack web development**.

The learning process is divided into 11 sequential, academic **Chapters**, ensuring a step-by-step understanding of how to build and scale these complex tools.

Summary of Resources and Techniques

Data Resources:

- **Optical & Thermal Sensors:** Landsat, Sentinel-2, MODIS.
- **Radar & Elevation:** Sentinel-1 SAR, SRTM / ALOS PALSAR DEMs.
- **Climatic Datasets:** WorldClim, CHIRPS.
- **In-Situ Data:** Local meteorological weather stations (e.g., CR2 Chile).

Advanced Analytical Techniques:

- **Agentic AI:** LangGraph for autonomous geospatial orchestration.
- **Deep Learning (PyTorch):** Multi-task CNNs for simultaneous lake, snow, and change detection segmentation.
- **Multi-Temporal Change Detection:** Time-series analysis across decades using fused SAR and Optical data.
- **Hydrological Modeling (ArcHydro / PySheds):** Watershed delineation and water balance.
- **Spatial Databases:** PostgreSQL + PostGIS for scalable vector storage.
- **Full-Stack Web GIS:** FastAPI + React + MapLibre GL JS for interactive dashboards.

Selected Region of Interest (ROI)

Torres del Paine National Park & Punta Arenas (Grey River Basin), Magallanes Region, Chile.

Detailed Project Structure and Academic Flow

CHAPTER 1: Climatic Variables & Image Processing

Academic Objective: Foundation in data acquisition, atmospheric correction, and climate modeling.

`Chapter_01/`

- `01_stac_multisensor_download.py`: (Concept: Programmatic acquisition). Uses Copernicus API and AWS STAC to download multi-sensor data.
- `02_atmospheric_correction.py`: (Concept: TOA to BOA reflectance physics). Radiometric/Atmospheric correction using Python (`rasterio`/`Py6S`) and ENVI.
- `03_station_ml_interpolation.py`: (Concept: ML for missing data). Training Random Forest on weather station data to predict continuous climate surfaces. **Includes automated Sensor Failure Anomaly Detection (`IsolationForest`) and manual visual QA/QC.**
- `04_precipitation_dual_analysis`: (Concept: R vs Python spatial analysis). CHIRPS historical anomalies and ArcGIS IsoCluster climatic zone classification. **Includes explicit R (`anomalize`) and ArcGIS Pro (Space-Time Pattern Mining) extreme climate event anomaly detection.**
- `05_uhi_modis_mapping.py`: (Concept: Urban microclimates). Urban Heat Island mapping over Punta Arenas using thermal data.

CHAPTER 2: Glacial Retreat, SAR Dynamics, and AI

Academic Objective: Tracking ice loss using multi-sensor approaches (Radar/Optical) and foundational Deep Learning.

`Chapter_02/`

- `01_glacier_area_perimeter.py`: (Concept: Vector GIS change detection). ArcGIS Pro/Python script to calculate historical Grey Glacier retreat.
- `02_sentinel1_sar_analysis.py`: (Concept: Cloud-penetrating radar physics). Classify dry snow vs. wet snow vs. ice using SAR backscatter.
- `03_xarray_ndwi_multitemporal.py`: (Concept: Spectral indices). Mapping Grey Lake expansion across decades.
- `04_pytorch_lake_segmentation.py`: (Concept: Semantic segmentation). U-Net CNN for automated glacial lake extraction.
- `05_arcpy_ndsi_snow_cover.py`: (Concept: Snow tracking). Automated ArcGIS Pro script for NDSI mapping.
- `06_multitemporal_change_detection.py`: (Concept: Multi-temporal sensor fusion). Fusing Sentinel-1 SAR and Sentinel-2 Optical time-series data to algorithmically detect land-cover changes and glacial calving events between two dates (T1 vs T2). **Includes ENVI RX Anomaly Detection and Python Time-Series analysis for predicting Glacial Lake Outburst Floods (GLOFs).**

CHAPTER 3: Hydrology and Watershed Modeling

Academic Objective: Understanding the mechanics of water flow, terrain modeling, and basin management.

`Chapter_03/`

- `01_dem_hydro_conditioning.py`: (Concept: Terrain correction). Filling sinks and applying TIN corrections to DEMs.
- `02_archydro_basin_delineation.py`: (Concept: Automated flow routing). Using ArcHydro and PySheds to delineate the Grey River Watershed.

- `03_morphometric_parameters.py`: (Concept: Basin physical traits). Calculate form factor, mean slope, and plot the Hypsometric Curve.
- `04_drainage_and_strahler.py`: (Concept: Stream hierarchy). Calculating drainage density and Strahler stream order.
- `05_water_balance_isohyets.py`: (Concept: Spatial hydrology inputs/outputs). Generate Isohyets and Isotherms for a spatial Water Balance model.

CHAPTER 4: Vulnerability and Ecosystem Impacts

Academic Objective: Understanding the biological and human consequences of the physical changes.

`Chapter_04/`

- `01_maxent_modeling` **(R/Python):** (Concept: Species Distribution Modeling). Map how flora/fauna will migrate as the watershed's climate shifts.
- `02_vulnerability_index_mce.py`: (Concept: Multi-Criteria Evaluation). Combine precipitation, glacial retreat, and flood risks into a "Climate Vulnerability Heatmap". **Includes Vegetation Stress/Burn Scar Anomaly Detection using R (`bfast`), ArcGIS Pro (CCDC), and manual visual interpretation.**

CHAPTER 5: Cartography and Management Plans

Academic Objective: Communicating spatial data clearly via standard GIS outputs.

`Chapter_05/`

- `01_map_automation_layout.py`: (Concept: Cartographic automation). `arcpy.mp` script generating standardized PDF maps.
- `02_watershed_management_report.py`: (Concept: Reporting). Compile stats, water balance, and vulnerability indices into a final text/PDF management plan.

CHAPTER 6: Capstone Project - The Linear Unified Pipeline

Academic Objective: Synthesize previous modules into a single, automated codebase.

`Chapter_06/`

- `main_linear_pipeline.py`: (Concept: System integration). A massive Python script that linearly triggers downloads, preprocessing, AI inference, watershed delineation, and report generation in one continuous execution loop.

ADVANCED ARCHITECTURES

CHAPTER 7: Agentic Orchestration (LangGraph)

Academic Objective: Move from rigid, linear scripting to autonomous, state-machine driven AI agents.

`Chapter_07/`

- **Concept:** Instead of a script running top-to-bottom, we build specialized AI agents that "decide" when to act based on the data state.
- **Action:** Build a LangGraph network with an `AcquisitionAgent`, `GlacierAgent`, `HydrologyAgent`, and `ReportAgent`. They autonomously pass data payloads (like GeoTIFFs) between each other, handling errors and retries intelligently.

CHAPTER 8: The Cascade Effect Modeling

Academic Objective: Move from isolated metrics to modeling a physical chain-reaction.

`Chapter_08/`

- **Concept:** Disasters don't happen in isolation. We will programmatically link the models to prove cause-and-effect.
- **Action:** A Python framework that specifically tracks the statistical causality: *Temperature Anomaly (Ch1) → Lake Expansion (Ch2) → Altered Water Balance (Ch3) → High Vulnerability Niche Shift (Ch4).* **Includes multi-platform validation against manually detected events.**

CHAPTER 9: Multi-Task PyTorch Model

Academic Objective: Upgrade standard Deep Learning to advanced multi-head architectures.

`Chapter_09/`

- **Concept:** Running multiple separate neural networks is computationally expensive. Multi-task learning predicts several things at once.
- **Action:** Upgrade the U-Net from Chapter 2 into a **Multi-Head CNN**. A single PyTorch model that ingests Optical, SAR, and DEM tensors simultaneously to output masks in one forward pass:
  - (1) Glacial Lake Segmentation
  - (2) Snow/Ice Cover
  - (3) Flood Risk Zones
  - **(4) Multi-Temporal Change Detection Head:** Anomaly scoring that highlights structural changes in the terrain between two different years.

CHAPTER 10: Spatial Database Integration (PostGIS)

Academic Objective: Move from flat files (Shapefiles/GeoTIFFs) to enterprise spatial databases.

`Chapter_10/`

- **Concept:** Flat files are hard to query over time. Databases allow for massive spatial SQL queries.
- **Action:** Set up a PostgreSQL + PostGIS database via Docker. Write Python (`SQLAlchemy`, `GeoAlchemy2`) scripts to ingest the output polygons (lakes, watersheds, vulnerability zones) and raster metadata directly into the database.

CHAPTER 11: Interactive Dashboard (Full-Stack GIS)

Academic Objective: Move from static PDF maps to real-time, interactive Web GIS.

`Chapter_11/`

- **Concept:** Stakeholders need interactive platforms, not just PDFs.
- **Action:** Build a **FastAPI** backend that queries the PostGIS database, and a **React + MapLibre GL JS** frontend to visualize the Climate Cascade, Glacial Retreat, and Watershed changes dynamically over a web browser.